

Sequence.

$$f(x) = x^2 - 5.$$

$$f'(x) = 2x$$

$$x_{n+1} = x_n - \frac{x_n^2 - 5}{2x_n}$$

Task 2

First from the recurrence relation, if $x_0 = 3$,

$$x_1 = x_0 - \frac{x_0^2 - 5}{2x_0}$$

$$= 3 - \frac{9 - 5}{6}$$

$$= 3 - \frac{4}{6} = 3 - \frac{2}{3}$$

$$= 7/3 \quad \checkmark.$$

Secondly, obtain from the geometry,
The tangent line has equation

$$y = mx + c \quad \text{where } m = f'(x_0)$$

$$= 2 \cdot 3 = 6.$$

The tangent line goes through

